

# $c = -2$ conformal field theory in quadratic band touching

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# Life in Hong Kong



**HKUST campus**



**Tropical fruit**

(Singular) QBT with two-component fermions:  $\hat{\psi} = (\hat{\psi}_1, \hat{\psi}_2)^T$ .

$$\hat{H} = \int \frac{d^2k}{(2\pi)^2} \hat{\psi}^\dagger(\mathbf{k}) H(\mathbf{k}) \hat{\psi}(\mathbf{k})$$

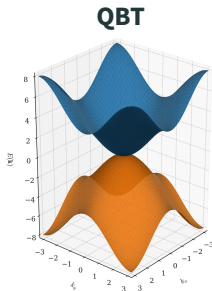
$$H(\mathbf{k}) = \begin{pmatrix} t_+ k_2^2 - t_- k_1^2 & (t_+ + t_-) k_1 k_2 \\ (t_+ + t_-) k_1 k_2 & t_+ k_1^2 - t_- k_2^2 \end{pmatrix}$$

$$\varepsilon_+(\mathbf{k}) = t_+ |\mathbf{k}|^2, \quad \varepsilon_-(\mathbf{k}) = -t_- |\mathbf{k}|^2$$

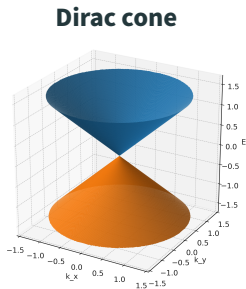
$t_+, t_- > 0$ ; half-filling.

# Quadratic Band Touching in 2 + 1D

QBT defines a **distinct universality class** ( $\leftrightarrow$  Dirac point).



$$\epsilon_{\pm} \propto \pm |\mathbf{k}|^2, \quad z = 2$$



$$\epsilon_{\pm} \propto \pm |\mathbf{k}|, \quad z = 1$$

Scaling:  $x \mapsto bx, \quad t \mapsto b^z t$

QBT appears in kagome, checkerboard, bilayer graphene, ...

[Liu, Yao, Ma PRB 2010; Sun PRB 2008; McCann & Fal'ko PRL 2006; ...]

**What is the characterization of QBT as a quantum critical point?**

# From Quantum Lifshitz to QBT

|         | Free fermion | Free boson              |
|---------|--------------|-------------------------|
| $z = 1$ | Dirac        | Klein–Gordon            |
| $z = 2$ | <b>QBT</b>   | <b>Quantum Lifshitz</b> |

**Entanglement on a torus:** QBT's subleading term is well fitted by the quantum Lifshitz scaling form. [Chen et al. 2015]

## Rokhsar–Kivelson ground state of the quantum Lifshitz model

$$\underbrace{\Psi_0[\phi]}_{\text{quantum amplitude}} = \underbrace{\sqrt{\frac{e^{-S_{2D}[\phi]}}{Z}}}_{\text{square root of classical probability}}$$

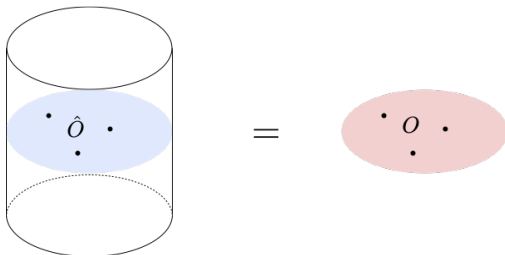
$$S_{2D}[\phi] = \frac{K}{2} \int d^2x (\nabla\phi)^2$$

$\phi(\mathbf{x})$ : scalar field;  $K$ : stiffness;  $Z$ : normalization.

[Ardonne, Fendley, Fradkin 2004]

## A 2D Theory for Equal-Time Correlations

In the quantum Lifshitz model, equal-time correlators are 2D classical averages.



$$\langle \Psi_0 | \hat{O}_1 \hat{O}_2 \hat{O}_3 | \Psi_0 \rangle = \frac{1}{Z} \int \mathcal{D}\phi O_1 O_2 O_3 e^{-S_{2D}[\phi]}.$$

$$\hat{O}_i = O[\hat{\phi}](\mathbf{x}_i), \quad O_i = O[\phi](\mathbf{x}_i) \quad (i = 1, 2, 3).$$

**What is the corresponding 2D theory for QBT?**

# Continuum Model

Fermions are represented as **1-forms**:

$$\hat{\psi}(\mathbf{x}) = \hat{\psi}_1(\mathbf{x})dx^1 + \hat{\psi}_2(\mathbf{x})dx^2$$

Nontrivial rotationally symmetric QBT Hamiltonian is given by [Fradkin (2013)]

$$\hat{H} = t_+ \int d\hat{\psi}^\dagger \wedge \star d\hat{\psi} + t_- \int \delta\hat{\psi} \wedge \star \delta\hat{\psi}^\dagger$$

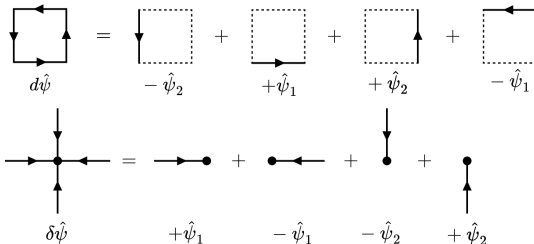
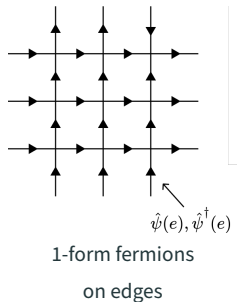
without chemical potential term (half-filling).

- $d\hat{\psi}$ : exterior derivative (curl)
- $\delta\hat{\psi} = -\star d\star\hat{\psi}$ : codifferential (divergence)

In momentum space:

$$H(\mathbf{k}) = \begin{pmatrix} t_+k_2^2 - t_-k_1^2 & (t_+ + t_-)k_1k_2 \\ (t_+ + t_-)k_1k_2 & t_+k_1^2 - t_-k_2^2 \end{pmatrix}$$

# Square Lattice Realization



$$\hat{H} = t_+ \sum_{f \in \text{Faces}} d\hat{\psi}^\dagger(f) d\hat{\psi}(f) + t_- \sum_{v \in \text{Vertices}} \delta\hat{\psi}(v) \delta\hat{\psi}^\dagger(v)$$

# Path Integral Representation of Ground State

A ground state has a **path integral representation** in terms of a Grassmann scalar  $\theta(x)$ :

$$|\text{GS}\rangle = \frac{1}{\sqrt{Z}} \int \theta_{\mathbf{k}=0} |\kappa d\theta\rangle \mathcal{D}\theta$$

- $Z$ : normalization
- $\kappa$ : const ( $= 1/\sqrt{4\pi}$  by convention)
- $|\kappa d\theta\rangle := e^{-\int \kappa d\theta \wedge \star \hat{\psi}^\dagger} |0\rangle$  (fermionic coherent state)

# Main Result

**Key step:** coherent state overlap produces the **symplectic fermion** action

$$\langle \kappa d\theta^* | \kappa d\theta \rangle = \exp \left( \kappa^2 \int d\theta^* \wedge \star d\theta \right) =: e^{-S[\theta, \theta^*]}$$

## Quantum-Classical Correspondence:

$$\begin{aligned} & \langle \text{GS} | F[\hat{\psi}^\dagger] G[\hat{\psi}] | \text{GS} \rangle \\ &= \frac{1}{Z} \int \mathcal{D}\theta \mathcal{D}\theta^* \theta_{\mathbf{k}=0}^* F[\kappa d\theta^*] G[\kappa d\theta] \theta_{\mathbf{k}=0} e^{-S[\theta, \theta^*]} \\ &= \langle \theta_{\mathbf{k}=0}^* F[\kappa d\theta^*] G[\kappa d\theta] \theta_{\mathbf{k}=0} \rangle_S \end{aligned}$$

Symplectic fermion theory is CFT  $\rightarrow$  Spatial conformal invariance!

2D symplectic fermion theory is a **logarithmic CFT with  $c = -2$**

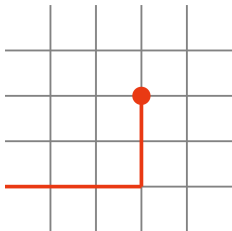
$$L_0 \begin{pmatrix} |A\rangle \\ |B\rangle \end{pmatrix} = \begin{pmatrix} h & 0 \\ \textcolor{red}{1} & h \end{pmatrix} \begin{pmatrix} |A\rangle \\ |B\rangle \end{pmatrix}$$

$\Downarrow$

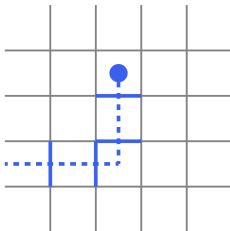
$$\langle A(0)A(x) \rangle = 0, \quad \langle A(0)B(x) \rangle = \frac{C}{|x|^{2h}}$$

$$\langle B(0)B(x) \rangle = \frac{D - \textcolor{red}{2}C \log |x|}{|x|^{2h}}$$

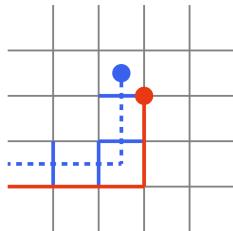
# Excitations: Topological Strings



$\theta$  / electric



$\phi^*$  / magnetic



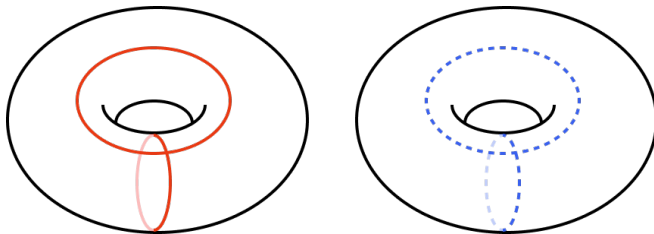
Composite

Excitations are created by **string operators**:

- $\theta$  (**electric**) excitation:  $|\theta(x_1)\theta(x_2)\rangle := \kappa^{-1} \int_{x_2}^{x_1} \hat{\psi} |\text{GS}\rangle$
- $\phi^*$  (**magnetic**) excitation:  $|\phi^*(\tilde{x}_1)\phi^*(\tilde{x}_2)\rangle := \kappa^{-1} \int_{\tilde{x}_2}^{\tilde{x}_1} \star \hat{\psi}^\dagger |\text{GS}\rangle$

Strings are **continuously deformable**.

# Topological Degeneracy



Non-contractible loops on a torus

Non-contractible string operators map  $|GS\rangle$  to other ground states

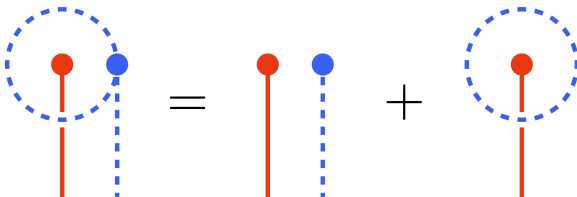
**$4^g$ -fold ground state degeneracy**  
on a genus- $g$  surface

Analogous to the toric code, but the system is *gapless*

# Non-diagonalizable Spin

Individual  $\theta$ ,  $\phi^*$  excitations have **trivial spin**: they return to themselves under  $2\pi$  rotation.

Rotate the composite  $\phi^*\theta$  excitation by  $2\pi$ :



$$R_{2\pi}(\phi^*\theta) = \phi^*\theta - 4\pi\mathbb{I}$$

$\phi^*\theta$  and  $\mathbb{I}$  form a **Jordan cell** under rotation  
the hallmark of logarithmic CFT

# Summary

- ✓ QBT ground state  $\Leftrightarrow$  **symplectic fermion theory**
- ✓ **Analogue of toric code**  
string operators,  $4^g$ -fold degeneracy, smooth/rough boundaries
- ✓ Composite excitation has **non-diagonalizable spin**

**arXiv:2511.16496**